Colloidal Hot-Injection Synthesis of $CuBSe_2$ Nanocrystals: Tetragonal Chalcogenide Templates for Superionic Lithiation Pathways

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Copper boron selenides represent a previously unexplored class of materials with tetragonal structural symmetries and bonding chemistries potentially relevant to lithium-ion conduction. In this work, the first colloidal synthesis of nanocrystalline copper boron selenides is reported, enabling access to nanocrystals with reaction times shown as the synthetic knob to control composition and morphology. Copper boron selenide nanocrystals are synthetically designed as precursors for lithiation reactions aimed at probing structural evolution and ionic mobility pathways toward superionic conductors applicable as solid electrolytes in lithium-ion batteries. Postsynthetic lithiation studies under ambient conditions reveal a series of phase transformations and morphological changes, which are characterized using a combination of X-ray diffraction and high-resolution transmission electron microscopy analysis. Thermometric and differential scanning calorimetry analyses on both copper boron selenide nanocrystals and their lithiated counterparts further uncover the associated signatures of superionic phase changes with lithium incorporation, suggesting dynamic structural rearrangements, and potential onset of fast ion transport. This study highlights colloidal synthesis as a powerful route to explore new composition spaces in complex chalcogenides and establishes copper boron selenides as a platform for investigating structure–transport relationships in emerging solid-state electrolyte chemistries.

1. Introduction

Solid-state electrolytes (SSEs) offer a transformative path toward safer and more energy-dense battery systems,[1–3] enabling the deployment of next-generation chemistries such as lithium-metal anodes[4] and high-voltage cathodes.[5] The synthetic breakthrough in crystalline fast-ion conductors of solid electrolytes has been slow progress with NASICON–$Li_{1+x}Al_xTi_{2-x}(PO_4)_3$ (LATP) and LAGP ($Li_{1+x}Al_xGe_{2-x}(PO_4)_3$),[6,7] garnet–$Li_7La_3Zr_2O_{12}$ (LLZO),[8] and sulfide–$Li_{10}GeP_2S_{12}$ (LGPS)[9]-based candidates discovered in the 2000s with promising electrochemical performance in the bulk form. Among the most promising candidates are superionic conductors[10–13] — materials that exhibit ionic conductivities (maximum values[14] are quantified at 10^{-2} S cm^{-1} at room temperature) approaching those of liquid electrolytes,[15] but with the inherent thermal and electrochemical stability of solids. Notable progress has been made in sulfide,[9] oxide,[10] and halide-based (Li_3YCl_6, Li_3InCl_6, Li_3ErI_6)[16,17] families,[18] and argyrodites (Li_6PS_5X[19] (X = Cl, Br, I)), with structural features such as interconnected conduction pathways[20,21] and dynamic sublattices[22] identified as key descriptors of superionic transport. However, traditional high-temperature solid-state synthesis routes or mechanical ball-milling processes often impose significant limitations on compositional tuning, defect engineering, lattice dynamics,[11] and particle morphology, which are critical parameters influencing interfacial compatibility and densification behavior in composite solid-state architectures.

In contrast, colloidal synthesis[23–27] has emerged as a versatile and chemically precise strategy for the bottom-up fabrication of nanostructured materials with well-defined composition, morphology, and surface chemistry. This approach leverages solution-phase control over nucleation[28] and growth to access metastable phases[29] and multicomponent systems[30] that are challenging to achieve via bulk methods. Recent work has demonstrated the extension of colloidal methods to complex chalcogenide[31] and halide[24] compositions, revealing new routes to compositionally tuned and size-controlled materials relevant to optoelectronic[32] and energy-storage[33] applications. In particular, surfactant-mediated synthetic routes[34–36] allow for the stabilization of low-dimensional or defect-rich phases, which may prove to be conducive to high[14] ionic conductivity, to the order

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of $\approx 1\,\mathrm{S\,cm^{-1}}$, in addition to a wide electrochemical voltage window or enhanced ionic mobility through engineered nanostructuring.[20]

Despite these advances, the application of colloidal chemistry to superionic conductors remains nascent. The potential to unlock previously inaccessible phase spaces, optimize ionic transport through morphology control, and tailor interparticle interfaces through ligand chemistry has not yet been fully realized in the context of SSEs. Recent studies[20,37–39] have highlighted this promise, demonstrating the colloidal synthesis of copper and lithium selenides, in particular, and sulfides[40] with controlled crystallinity and dopant incorporation. However, a systematic strategy to design, synthesize, and characterize superionic conductors via colloidal routes has yet to be established.

Here, we report on a colloidal synthesis platform for accessing nanocrystalline superionic conductors with tunable structure and morphology. By adapting principles from semiconductor nanocrystal (NC) synthesis to copper-containing chalcogenide systems, we establish a robust synthetic route to phase-targeted, size-controlled NCs that potentially exhibit high room-temperature ionic conductivity. The rational design of these NC compositions was based on the conclusions from a recent study combining high-throughput screening approach[41] for identifying promising materials using a combination of bond-valence methods and graph neural networks. In particular, the $LiBSe_2$ composition with an estimated ionic conductivity of $0.26\,\mathrm{mS\,cm^{-1}}$[41] was one of the top ten promising solid electrolyte candidates as identified from the high-throughput screening study. Since direct synthesis of size-controlled lithium-containing chalcogenide NCs is still an ongoing work, we instead synthesize the copper boron selenides through direct synthesis and study the postsynthetic structural and morphological changes upon lithiation. Another challenge with direct syntheses of lithium containing materials at high temperatures typically required in inorganic colloidal chemistry is the possibility of corrosion to the glassware. Structural studies combining X-ray diffraction (XRD) and high-resolution transmission electron microscopy (HRTEM) reveal the formation of compositionally complex metastable phases stabilized at the nanoscale. These results demonstrate that colloidal chemistry offers a powerful new axis for the design of superionic conductors, enabling integration of structural complexity and interfacial tunability in a solution-processable form.

2. Results and Discussions

2.1. Direct Syntheses of Copper Boron Selenide NCs Using Colloidal Hot-Injection Method

$CuBSe_2$ NCs with various morphologies (Figure S1, Supporting Information) were synthetized using a colloidal hot-injection synthesis technique with the procedure developed by our group. A detailed description of the synthetic procedure (**Figure 1**a) is provided in the methods. The colloidal synthesis of copper boron selenides enables precise control over NC nucleation and growth dynamics, offering a pathway to access novel compositions and morphologies that are challenging to obtain via traditional solid-state methods. In this system, the reactivity of the copper (copper acetylacetonate and copper chloride were tested) and boron (boric acid and trimethyl boron were tested) precursors, along with the nature of the selenium source (e.g., trioctylphosphine selenide (TOP-Se), Se in octadecene, direct Se injection were tested), plays a central role in dictating the kinetics of nucleation and the subsequent evolution of particle shape and size. Briefly, copper (copper chloride was demonstrated to be the most efficient copper precursor) and boron (boric acid) precursors were loaded in a round-bottom flask along with organic ligands and solvent and degassed before being placed at elevated temperatures. The potentially optimal lower limit of the reaction temperature was found out through an empirical optimization method, with more details in the Experimental Section. The selenium precursor (Se in octadecane) was injected, and the reaction times were varied between 20 and 120 mins to evaluate the role as an independent parameter in the ensuing structure–property studies with these NCs. The NCs were purified twice, according to the protocol mentioned in the Experimental Section, and stored in toluene for subsequent experimentation under ambient conditions in air. The starting amount of the $CuBSe_2$ NCs was 940 mM in 3 mL total of toluene, as calculated from the NCs dried from known amounts of dispersions. XRD experiments were performed using grazing-incidence angle with CuK_α radiation to elucidate the structural information of these NCs using colloidal dispersions drop cast on Si substrates. Patterns from all four reaction times are displayed in Figure 1b between 20 till 90° 2θ alongside reference patterns. The 20 and 120 min syntheses samples depict larger grain boundaries, as is evident from the lack of peak broadening compared to the 40 min and 60 min samples. These patterns were matched with $CuBSe_2$ (JCPDS card 04-024-5336) with primary reflections at 26.6° (112), 44° (204); CuSe (JCPDS 01-086-1240), $CuSe_2$ (JCPDS 04-004-2178), B_2O_3 (JCPDS 04-007-1547), and $Cu(OH)_2$ (JCPDS 04-009-4366), respectively as shown in Figure 1b.

The primary phase observed in the $CuBSe_2$ with body-centered tetragonal symmetry and *I-42 d* (122) space group is visualized in Figure S3, Supporting Information. B atoms are placed in *2a*, Cu atoms are placed exclusively in the *2b*, and Se atoms are placed in the *4c* positions (lattice constant: 5.78 Å for the sample with 40 min reaction time). Our copper boron selenides are a sacrificial template to prepare lithiated versions with lithium boron selenide through cation exchange.[37,42,43] Hence, we look into the cationic site occupancy behavior in $CuBSe_2$ to determine the viability of superionic conduction. The cationic site occupancy in tetragonal symmetry, generally speaking, can have a major effect on Li^+ ion conductivity. In fast-ion conductors such as Li superionic conductors (LISICONs) or $Li_7La_3Zr_2O_{12}$ (LLZO)-type garnets,[8] superionic phase possess a strong correspondence to distorted or tetragonal-like symmetry, with lithium present in the *8a*, *16f*, and *32g* nonequivalent Wyckoff positions. Lithium can be placed in either tetrahedral or octahedral sites, with coordination of four and six, respectively, although the migration path energies[15] associated with lithium motion are very different for these sites. With the copper boron selenides synthesized in a tetragonal structure suitable for superionic conduction, our goal is to investigate whether intrinsic defect processes can be induced upon lithiation, and if we can rationally implement the replacement of copper with lithium in interstitial positions, feasible toward superionic conduction. The Frenkel defect formation energy of lithium is significantly lower than that of cation

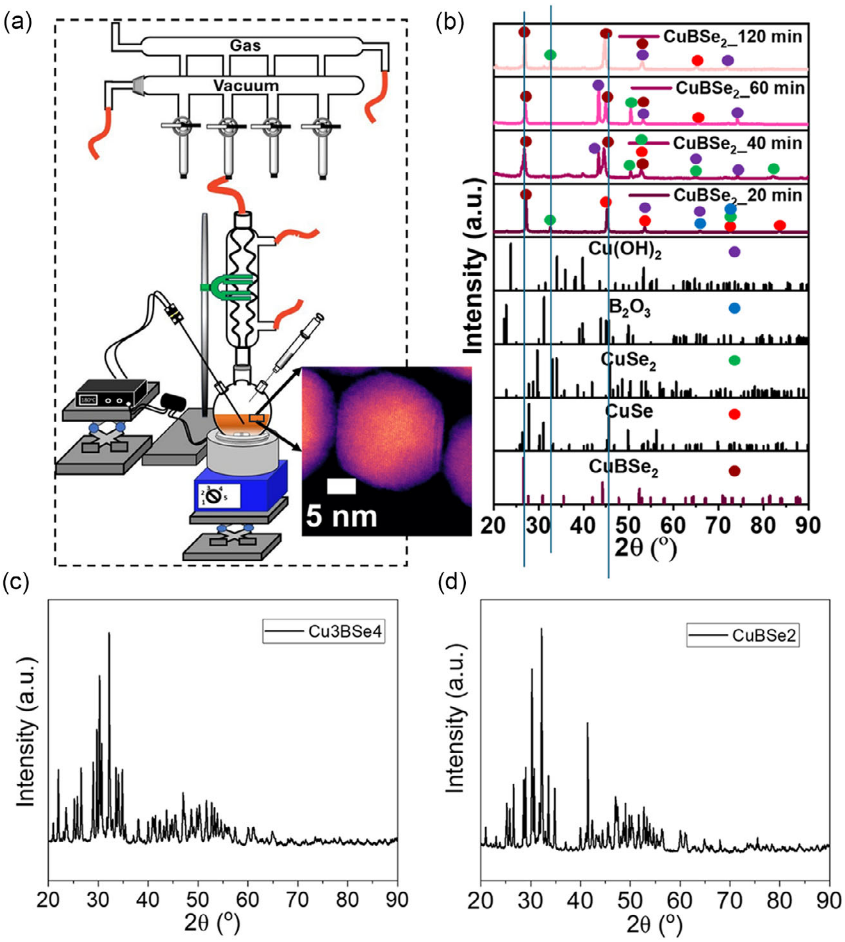

Figure 1. a) Hot colloidal injection setup using a Schlenk-line with reaction temperature, time, and precursors being variable parameters. b) XRD indicating the identification of $CuBSe_2$, CuSe, $CuSe_2$, B_2O_3, and $Cu(OH)_2$ phases in various stoichiometries which varied with samples based on reaction times. (Inset) The scanning transmission electron microscopes image of a representative $CuBSe_2$ NC obtained with 40 min reaction time with 5 nm scale bar. c-d) Solid-state syntheses of complex chalcogenides with stoichiometries of precursors set as Cu_3BSe_4 and $CuBSe_2$, respectively.

Frenkel reactions,[8] upon replacement of copper partially with lithium, suggesting that lithium is likely to occupy to be placed in interstitial positions with lower activation energies.

To further ascertain the need for the development of colloidal hot-injection synthesis routes, XRD data is presented in Figure 1c, d to showcase the absence of phase pure products obtained via a solid-state synthesis route. The Cu, B, and Se precursors for solid-state route are mentioned in the Experimental Section, and the reaction conditions are quite similar to the rest of our studies. However, we note that solid-state routes give rise to multiple phases, which are difficult to segregate even on purification cycles.

2.2. Morphological Variation in $CuBSe_2$ NCs with Synthetic Reaction Time and their Effect on Thermometry Characterization

It was shown previously that NC size plays an important role in the stabilization of crystal symmetries,[20] leading to superionic conduction as a function of temperature. Therefore, we next used transmission electron microscopy (TEM) studies to image and quantify the morphological features of these $CuBSe_2$ NCs. Samples obtained upon direct synthesis were purified with anti-solvents, as detailed in the Experimental Section to remove excess surface ligands in the form of amines and were drop cast on ultrathin microscopy grids to be imaged. The images from representative particles from each synthetic reaction time, shown in **Figure 2**a–d indicate that the 60 min (Figure 2c) reaction time yields the smallest size followed by 40 min time(Figure 2b). The absence of noticeable broadening in the diffraction peaks in the 20 and 120 min reaction samples from Figure 1b is reflected in the larger grain sizes shown in Figure 2a,d. Interestingly, the NC growth pattern is nonlinear with respect to reaction time. Detailed size distribution analyses on these particles are shown in Figure S2, Supporting Information

Upon injection of the selenium precursor into a hot coordinating solvent containing metal precursors and ligands, a burst nucleation event[44] is followed by a diffusion-controlled growth regime. In this LaMer-type[45] mechanism, cluster size distribution resulting from nucleation is modulated by parameters such

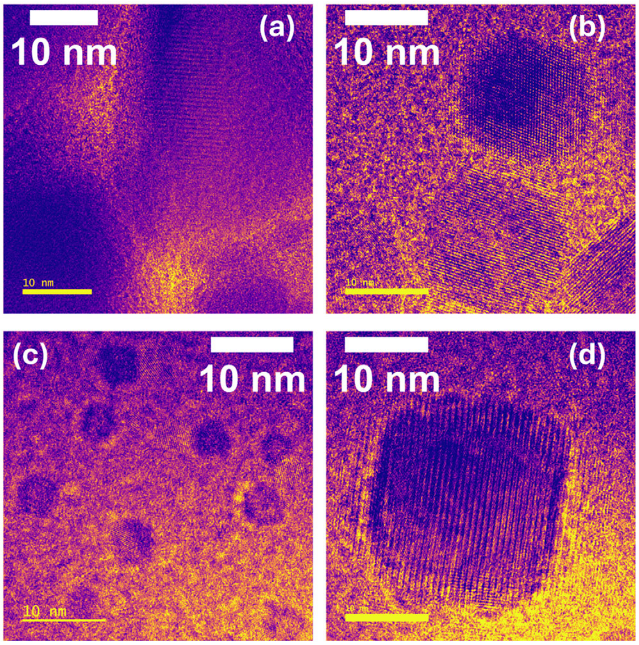

Figure 2. HRTEM images of $CuBSe_2$ NCs synthetized with reaction times of a) 20 min, b) 40 min, c) 60 min, and d) 120 min, respectively. Scale bars are 10 nm each.

as precursor concentration, ligand identity, and reaction temperature. With the choice of metal or selenide precursors other than the particular combination used in the Experimental Section, we were unsuccessful in obtaining the $CuBSe_2$ composition. Notably, reaction time-dependent studies reveal that the copper boron selenide NCs undergo significant morphological evolution as a function of reaction duration: early-stage products with 20 min reaction time typically exhibit burst nucleation followed by rapid aggregation, followed by faceted NCs at 40 min time, followed by small, quasispherical morphologies (5–10 nm) with 60 min time, while prolonged reaction times lead to larger, more faceted NCs with isotropic features such as hexagonal nanoplates.

This shape and size evolution can be attributed to a combination of Ostwald ripening and oriented attachment mechanisms.[46] In particular, the dynamic exchange between surface-bound ligands and metal–chalcogenide complexes facilitates anisotropic growth[47] by selectively passivating specific crystal facets, leading to shape anisotropy over time. The incorporation of boron introduces additional complexity due to its coordination chemistry and potential influence on local bonding networks within the growing crystal lattice. Such effects have been noted in related complex chalcogenide systems,[48] where heterometallic interactions and precursor reactivity imbalance contribute to phase evolution and facet-selective growth.

Together, these findings suggest that copper boron chalcogenide NC growth proceeds via kinetically controlled pathways that are tunable through reaction time and synthetic parameters. A mechanistic hypothesis of the formation of the copper boron selenide NCs, that is well-substantiated through diffraction patterns, is that the synthesis proceeds through a two-step mechanism. The formation of binary $Cu_{2-x}Se$ NCs (Step 1) is typically succeeded with the incorporation of boron to produce the ternary $CuBSe_2$ NCs (Step 2). For Step 1, the CuCl is dissolved in ODE, often with coordinating ligands like oleylamine (OLA) to stabilize Cu^+ ions and act as the copper precursor activator. Elemental selenium is dissolved in OLA or trioctylphosphine (TOP) to form a reactive Se precursor. The Se precursor is swiftly injected into the hot Cu-containing solution (at 310°C in our experiments), leading to the rapid formation of $Cu_{2-x}Se$ NCs. The first step can be demonstrated as

$$2CuCl + Se \rightarrow Cu_2Se + 2Cl^- \quad (1)$$

At elevated temperatures, boric acid decomposes to form boron oxide species, such as B_2O_3, as substantiated through our XRD experiments.

$$2\,H_3BO_3 \rightarrow B_2O_3 + 3H_2O \quad (2)$$

The boron oxide species interact with the surface or lattice of the preformed $Cu_{2-x}Se$ nanocrystals, leading to the formation of ternary $CuBSe_2$ NCs.

$$Cu_2Se + B_2O_3 \rightarrow 2CuBSe_2 \quad (3)$$

These insights lay the groundwork for rational design of morphology-engineered precursors for downstream lithiation and ion transport studies. We investigated the potential implication of average NC sizes with the existence of crystal symmetries leading to the stabilization of superionic conductivity using differential scanning calorimetry (DSC) measurements in **Figure 3**. The cooling and heating curves were run twice to investigate whether any features arising in the spectra were originating from the samples, or the ligands, and whether there were multiple crystallization processes. The 20, 40, and 120 min samples in Figure 3a,b,d are featureless, with clear lack of endothermic peaks indicative of NC melting, nor any exothermic peaks indicative of crystalline rearrangements in this temperature range. Interestingly, copper selenides[20] with sizes similar to the 40 min sample exhibit superionic to nonsuperionic phase peak around 133 °C, whereas the sizes closer to our 60 min sample show a peak at significantly lower temperatures of around 2 °C, respectively. The 60 min samples of copper boron selenides in this study, shown in Figure 3c, with median NC diameter of 7.6 nm show weak endothermic/exothermic peaks during both heating/cooling cycles near 9.3/4.6 °C. The presence of a population of larger particles, as an unintended side effect of synthesis in the 60 min sample, which are typically not all removed through size-selective centrifugation, could be responsible for the relatively weak endothermic/exothermic peaks in this sample, arising from the percentage of NCs with smaller sizes. The presence of these peaks points to the 60 min $CuBSe_2$ NCs existing in a potentially superionic state, although electrochemical impedance spectroscopy measurements will need to be performed to understand the ionic transport properties below and above the 9.3/4.6 °C in more detail in a future study. We also note the presence of similar features in prior reports on size-dependent superionic behavior in copper selenide[20] NCs. Although similar features could arise from ligand desorption or residual solvent evaporation, these will be expected to take place at significantly higher temperatures, i.e., between

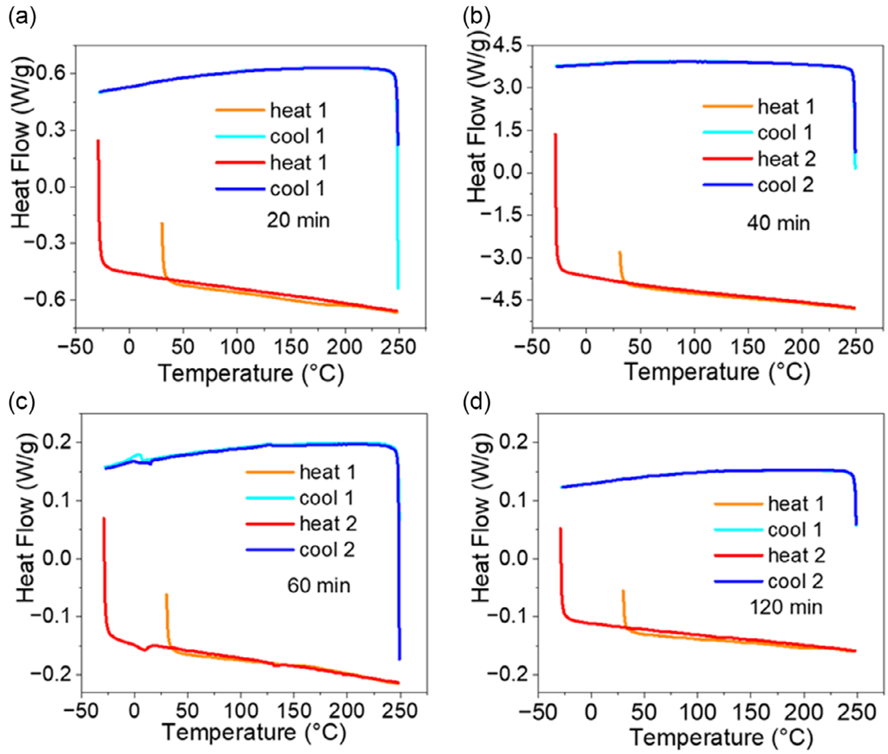

Figure 3. DSC measurements of $CuBSe_2$ NCs with various synthetic reaction times with two consecutive heating and cooling cycles between −30 and 250 °C. a) 20 min sample, b) 40 min, and d) 120 min samples are featureless sin this range, whereas the c) 60 min sample shows weak peak near ambient temperatures.

200–500 °C for oleylamine[49] and above 80 °C for toluene,[50] respectively.

2.3. Postsynthetic Cation Exchange Reaction (CER) with Lithium Precursors to Create Lithiated CuBSe2 Nanostructures (NSs)

We explored the ambient-temperature lithiation of $CuBSe_2$ NCs, prepared via postsynthetic lithiation, with tetragonal crystal symmetries as the starting structures. The progression of lithiation was tracked with TEM studies (Figure S4, Supporting Information), with the final structures presented in **Figure 4**. Compared to the relatively smaller NC sizes obtained with the (Figure 4a) 40 min and (Figure 4c) 60 min reaction times, the lithiated versions produce nanorods (Figure 4b) and nanodiscs (Figure 4d), respectively. Furthermore, although the lithiated samples with 20 min reaction (Figure 4e) do not exhibit well-defined morphologies, the 120 min sample presents triangular nanosheets (Figure 4f). The tunable morphologies can be potentially attributed to the presence of various secondary phases in variable amounts as depicted in Figure 4g. We found significant $CuBSe_2$ (JCPDS 04-024-5336) that was residual in these nanostructures, alongside the $LiBSe_2$ (JCPDS 04-025-6857), Li(OH) (JCPDS 00-032-0564), Li_2O (JCPDS 01-086-3880), and $LiBO_2$ (JCPDS 00-047-0340) phases. The $LiBSe_2$ phase, which provided the closest match with these structures, possesses monoclinic symmetry I(2/a) space group 15 with lithium in the *8f* and *4e* positions shown in Figure S5, Supporting Information. The monoclinic phase in superionic literature is observed with argyrodites,[19] thiophosphates, and oxides.[51] Lithiation leading to NC growth has been observed before with copper selenides,[37] with the mean NC diameter increasing from 6.1 to 7.8 nm as a result of diffusion-driven restructuring. HRTEM analysis reveals distinct morphological and structural transformations associated with the lithiation of $CuBSe_2$ NCs, depending on reaction time. In the 40 min sample in **Figure 5**a–c, a thick nanorod is observed embedded within a matrix of smaller NCs that resemble the morphology of the unlithiated sample. Upon closer inspection (Figure 4b,c), these smaller NCs retain discrete crystalline identities, suggesting incomplete lithiation and/or a kinetically arrested lithiation front.[52] This observation aligns with previous findings in nanoscale systems where lithiation often proceeds heterogeneously[53,54] due to diffusion limitations and surface energy disparities among particles of varying sizes.

The large nanorod—presumably more extensively lithiated—exhibits a complex internal structure with multiple crystallite domains extending across the particle. This implies that lithiation induces significant atomic reorganization and perhaps local recrystallization within individual NCs, similar to reports[55] of phase boundary migration and domain realignment observed in other multinary chalcogenides undergoing ion exchange or electrochemical lithiation. Such texturing may arise from the interplay between strain accommodation and lithium intercalation pathways, further modulated by crystal anisotropy and facet-selective reactivity.[56]

After lithiation of the 60 min sample (Figure 5d–f), the NC ensemble exhibits a notable shift in morphology, with signs of

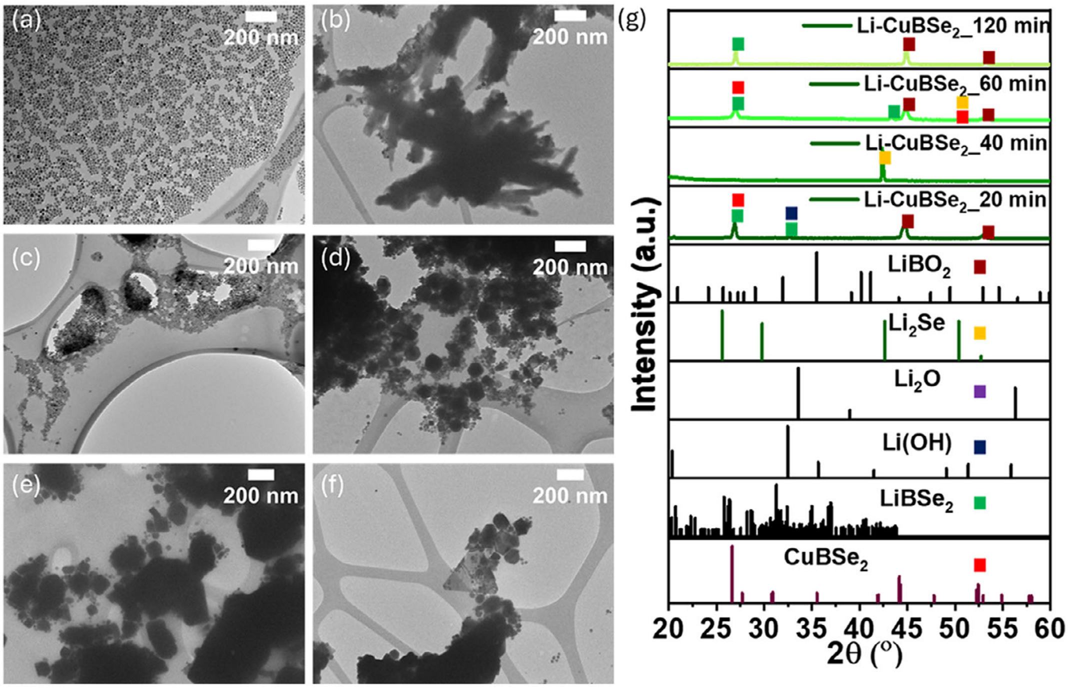

Figure 4. Progression of room-temperature lithiation in $CuBSe_2$ NCs synthesized at various reaction times leads to tunable morphologies. The $CuBSe_2$ NCs synthesized with 40 min time in a) lead to nanorod like structures in b) the 60 in sample c) with the smallest average NC size and upon lithiation leads to flatter nanodiscs in d) the lithiated samples from 20 min samples in e) and do not present particularly define morphologies. However, the lithiated samples with 120 min reaction time in f) lead to triangular nanosheets. g) XRD patterns obtained on the lithiated samples show the presence of remnant $CuBSe_2$ with the emergence of $LiBSe_2$, Li(OH), Li_2O, and $LiBO_2$ phases in various ratios potentially leading to various morphologies.

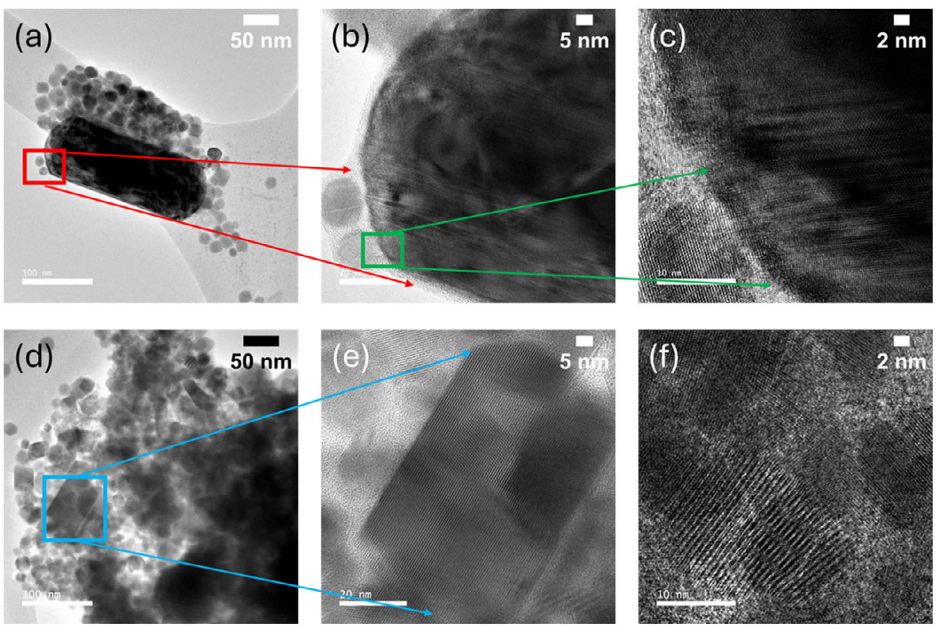

Figure 5. HRTEM images with various magnifications used to understand the domains of lithiated $CuBSe_2$ NCs with (a–c) 40 min and (d–f) 60 min reaction times, respectively. The thicker nanorod in a) surrounded by smaller NCs reminiscent of 40 min unlithiated sample is imaged with higher magnification focusing at the surface in b,c) which shows the smaller NCs be to distinct identities. The larger nanorod, presumably lithiated is shown to possess distinct crystallite directions which continue through the larger NC. The lithiated 60 min sample in d) shows changes in morphology throughout the populations with Oswald ripening-like behavior, the magnified images in e,f) show the smaller NCs to grow and form the larger faceted nanosheet. The scale bars are 50 nm (a,d), 5 nm (b,e), and 2 nm (c,f), respectively.

Ostwald ripening and the emergence of larger, faceted nanosheets. These morphological features suggest a dynamic coalescence and recrystallization process, likely driven by the chemical potential gradients introduced by lithiation.[57] The high-resolution images (Figure 5 panels e and f) reveal that formerly discrete NCs appear to fuse and reorganize, indicating a progressive phase transformation toward more thermodynamically stable morphologies, consistent with prior studies on solution-mediated ripening in lithiated semiconductor NCs.[58,59]

Detailed Rietveld analyses of the diffraction patterns from copper boron selenide NCs and their lithiated counterpart revealed the tetragonal *fcc* (space group 122 with symmetry *I*-42 *d*) to centrosymmetric monoclinic (space group 15 with symmetry *C*2/*c*) transformations upon chemical lithiation. Table S2 and S3, Supporting Information, present the detailed information from the refinement tables from the samples categorized with their composition and syntheses times. Figure S4 through S13, Supporting Information, depict the visualization of crystal symmetries and crystallographic information from various reaction times as obtained upon refinement analyses.

The DSC results show a prominent endothermic peak at 254 °C that is universal in all samples (**Figure 6**a–d), which is indicative of the melting of residual lithium nitrate[60] precursors. This endothermic event is consistently observed in all lithiated samples and is assigned to the melting of unreacted $LiNO_3$ used in the postsynthetic lithiation process. In the 20 min lithiated

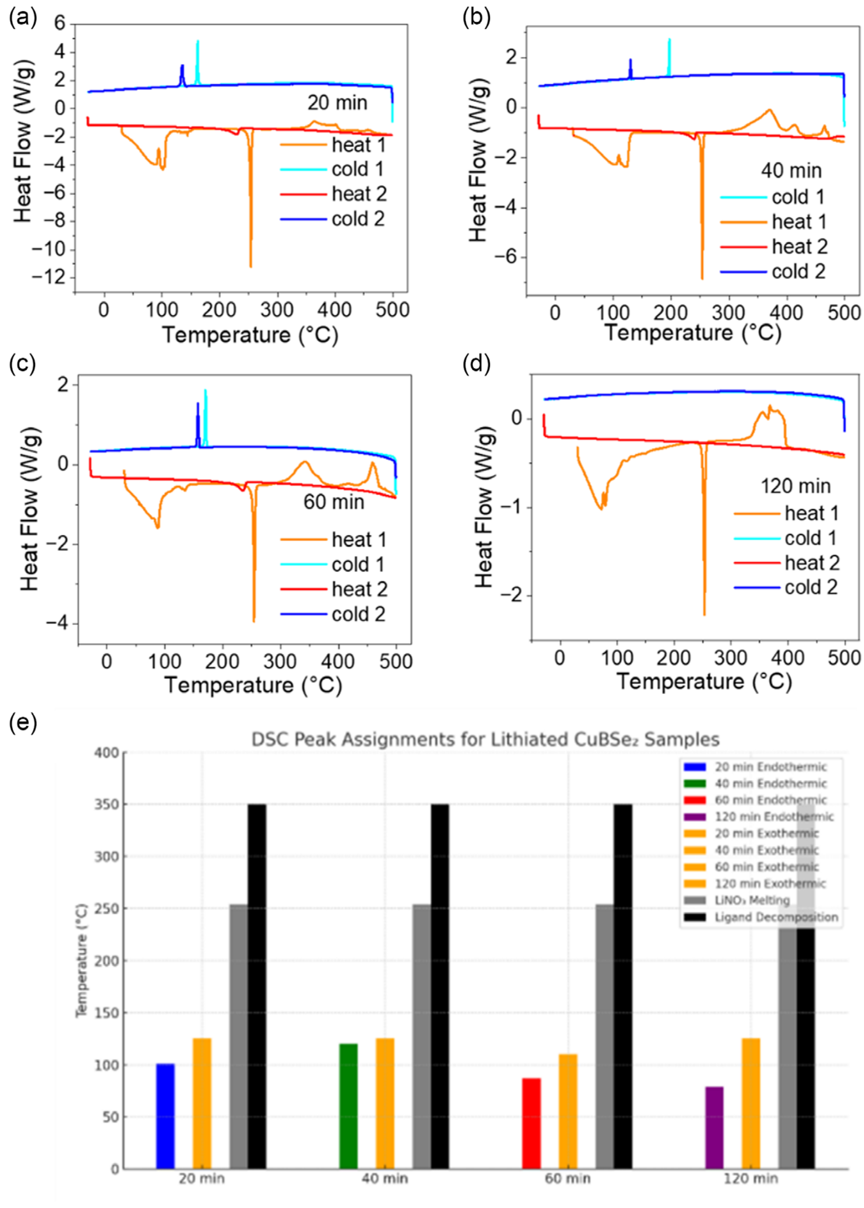

Figure 6. The DSC measurements with lithiated samples formed from $CuBSe_2$ NCs with reaction times of a) 20 min, b) 40 min, c) 60 min, and d) 120 min are shown. Two cooling and heating cycles were performed consecutively between −30 to 500 °C. e) DSC peak assignments for lithiated $CuBSe_2$ NC samples.

sample, endothermic peaks at ≈84 and ≈101 °C are assigned to phase transformations of nanostructured Li_2Se and $LiBSe_2$. These peaks align with prior reports of size-dependent superionic phase transitions in lithiated copper selenides around ≈68 °C for NCs ≈7.8 nm, supporting the hypothesis that similar nanoscale effects shift the superionic transition in lithium boron selenides. It was previously demonstrated that endothermic peaks indicative of superionic conduction in nanostructured lithium selenides could be observed using DSC thermogram centered around 68 °C for NCs with median diameter if 7.8 nm,[37] although this composition does not feature such peaks in its bulk form. The presence of two peaks may arise from particle size polydispersity, which was somewhat difficult to control in lithiated samples. The exothermic peaks centered around 110–125 °C in the second cooling curves indicate crystalline rearrangements, possibly monoclinic ↔ tetragonal/rhombohedral transitions. These suggest reversible structural transitions in $LiBSe_2$ related to lithium sublattice disordering typical of superionic phase transitions. The broader exothermic peaks centered around 350 °C in the cooling curves which show up in all samples could be attributed to the decomposition of the oleyl amine surface ligands[61] in these materials bound to the nanostructures in various amounts.

In the 40 min lithiated sample, the endothermic peaks at ≈101 and ≈120 °C are assigned to superionic transition(s) in nanostructured $LiBSe_2$ variants. These peaks are slightly higher than those in the 20 min sample, suggesting increased stability of the superionic phase with more defined nanorod morphology. Two-phase rearrangement in the exothermic peaks upon cooling observed suggests possible coexistence of different structural domains due to incomplete lithiation or morphology-induced heterogeneity. Ligand decomposition is assigned to a peak at 350 °C similar to the sample in the previous paragraph.

The exothermic peak in the 60 min lithiated sample is assigned to superionic transition in uniformly sized $LiBSe_2$ nanodiscs. This sample exhibits a single, sharper peak suggesting improved crystallinity and reduced size dispersion, consistent with monodisperse NCs. Exothermic peaks (cooling) nearly overlap and indicate a single-phase transformation, consistent with more uniform lithiation and crystallographic texture, as corroborated by TEM (Figure 5d–f). In the 120 min lithiated sample, endothermic peaks at ≈70 and ≈79 °C likely arise from superionic transitions in larger triangular nanosheets with greater size distribution and multiple Li-containing phases. The exothermic peaks observed upon cooling demonstrate multiple transitions. These can be assigned to polymorphic transitions potentially due to higher crystalline complexity and coexisting phases such as B_2O_3, $LiBO_2$, and Li_2O (Figure 4g). While some of these impurities may impede long-range transport, others could facilitate short-range interfacial conduction. These results have been summarized in **Table 1**.

While these features are suggestive of temperature-dependent lithium mobility, we emphasize that DSC provides indirect evidence. Additional measurements—such as variable-temperature impedance spectroscopy, Raman spectroscopy, and XRD—are essential to confirm superionic conduction. These studies are ongoing and will be reported in future work.

Table 1. The key thermal features and their tentative assignments across all lithiated samples as visualized in **Figure 6**e.

Feature	Temperature [°C]	Indicative Assignments	Reference
Endothermic	≈70–120	Superionic transitions in $LiBSe_2$	Banerjee & Jain[37]
Exothermic	≈110–125	Crystalline phase rearrangements	Kravchenko & Sigaryov[51]
Endothermic	≈254	Residual $LiNO_3$ melting	Ruiz et al.[60]
Exothermic	≈350	Oleylamine ligand decomposition	Chang et al.[61]

3. Conclusions

In this work, we have demonstrated the successful colloidal hot-injection synthesis of copper boron selenide ($CuBSe_2$) NCs as a new class of chalcogenide-based precursors for exploring superionic conduction via postsynthetic lithiation. By systematically varying reaction time, we achieved control over NC size and morphology, revealing nonlinear growth behavior influenced by nucleation and ripening kinetics. Structural and thermometry characterization through XRD, HRTEM, and DSC confirmed the formation of compositionally complex and size-tunable phases, with tetragonal $CuBSe_2$ emerging as the primary phase under optimized conditions.

Postsynthetic lithiation under ambient conditions induced substantial morphological and structural transformations in $CuBSe_2$ NCs. Lithiation of smaller NCs obtained from intermediate reaction times (40 and 60 mins) led to the formation of distinct nanorods and nanosheets, while HRTEM revealed domain formation, crystallite reorientation, and Ostwald ripening behaviors. These observations underscore the role of particle size and morphology in determining lithiation dynamics and resulting nanostructure architectures.

Thermometric analyses using DSC provided preliminary evidence for temperature-dependent phase behavior in both pristine and lithiated $CuBSe_2$ NCs. In particular, the 60 min sample displayed thermal signatures suggesting structural rearrangements potentially linked to superionic phases. These findings lay the groundwork for future electrochemical studies aimed at probing ionic conductivity in this emerging material system. Lithiated samples showed more complex phase behavior, indicative of crystalline rearrangement potentially between monoclinic, tetragonal, or rhombohedral symmetries. The DSC data for lithiated $CuBSe_2$ nanostructures reveal size- and morphology-dependent thermodynamic signatures consistent with superionic phase transitions, particularly in the 60 min sample. These features, especially the sharp, reproducible transitions near ambient temperatures suggest promising temperature-driven ionic transport behavior. Further validation via extensive temperature-dependent impedance spectroscopy is needed to definitively link these thermal events to ionic conductivity. Computational studies using atomistic simulations are to be reported in a future companion article combined with extended datasets containing multiple compositions in an attempt to generalize the mechanistic interpretation of our findings. As a first report focused on the colloidal synthesis and lithiation-induced phase/morphology evolution of $CuBSe_2$, our current work does not include battery integration or electrochemical cycling studies. These efforts are indeed essential and are part of ongoing work in our laboratory which will be focused on electrochemical testing, including Li metal compatibility and cycling, which will be addressed in future studies. A fully systematic design-of-experiments (DOE) optimization is an important future direction.

This study highlights the power of colloidal synthesis in accessing metastable, compositionally diverse chalcogenide phases with tunable NC size and morphology, which are critical parameters for enabling superionic behavior. Our results establish $CuBSe_2$ as a promising precursor framework for the rational design of nanostructured solid electrolytes and open new avenues for integrating bottom-up chemistry into solid-state battery materials discovery.

4. Experimental Section

Colloidal Hot-Injection Synthesis of $CuBSe_2$: 2 mmol (0.198 g) of anhydrous CuCl and 2 mmol (0.124 g) of H_3BO_3 were dissolved in a solvent mixture of 10 mL of oleylamine (OLA) and 10 mL of 1-octadecane (ODE) in a 50 mL round-bottom flask. This precursor mixture containing copper and boron salts was heated to a temperature for 80 °C and subject to vacuum degassing for 2 h. Reduction of the degassing time led to segregation of copper during the purification procedure as determined by the blue color of the supernatant. The reaction temperature was set to be raised to 310 °C using a J-KEM dual-temperature Gemini-2 controller under nitrogen (N_2) flow and glass wool was used to isolate the system for thermostability. This reaction temperature was chosen since it was observed the synthesis was able to recover to this temperature upon Se injection in a reproducible manner. We tried to set the reaction temperature in a range between 280 and 350 °C; however, the lower temperatures did not lead to NC formation, and the higher temperatures were not approached even after a period of hours following the Se injection. The empirically determined lower temperature limit was found to be ≈280 °C, below which NC formation was not observed. The effect of reaction recovery temperature (≈310 °C) after selenium precursor injection was a byproduct of the various volumes of precursors added during the synthesis, and this value was used as a set point to normalize the recovery times across various synthesis reported in this study. Reactions at 350 °C were not achievable postinjection due to thermal inertia. The Se-ODE solution was prepared in the glove box by dissolving 10 mmol of Se in 10 mL of ODE and heating the mixture to 200 °C for 40 min under stirring and then sonicating for 10 min until the Se was dissolved. Once the temperature was stabilized at 320.7 °C under N_2 flow, 4 mmol (0.5 M) of the Se-ODE solution was rapidly injected using a 16-gauge syringe into the reaction flask. The temperature dropped with the addition of Se-ODE. Once the reaction temperature reached 310 °C, the reaction was allowed to proceed for 20, 40, 60, and 120 min, respectively in various batches to facilitate NC growth. Afterward, the reaction mixture was allowed to cool to room temperature before the addition of 10 mL of toluene. Then 5 mL of ethanol was added to the reaction mixture and centrifuged for 5 mins at 6000 rpm for removal of unreacted precursors. The NCs obtained as the precipitate were redispersed in toluene for further characterization.

Solid-State Syntheses of Cu–B–Se Stoichiometries: Solid-state syntheses with Cu–B–Se compositions were attempted with stoichiometric amounts of solid precursors (copper chloride, boric acid, and selenium) for Cu_3BSe_4 and $CuBSe_2$ compositions. After degassing the precursor combination at 100 °C for 2 h, the reaction temperature was raised to 300 °C under N_2 flow and held there for various amount of reaction times ranging between 20 min till 120 min. Upon quenching the reaction these materials were recovered in powder form and washed two times with methanol using centrifugation and dried afterwards.

Lithiation to Li–B–Se Compositions: For ambient-temperature lithiation reactions with the $CuBSe_2$ NCs, ≈942 mM of LiNO3 in 5 mL methanol was prepared as the precursor stock solution. After reacting between ≈3 h to several days we did not notice any difference in lithium content in the sample using XRD. Hence ≈3 h of reaction time was used in subsequent reactions. Upon completion of the reaction the solution was sonicated for 10 min, 5 mL methanol was added followed by centrifugation at 6000 rpm for 10 min to remove the unreacted precursors. The NCs obtained as the precipitate were redispersed in toluene for further characterization.

XRD Experiments: XRD experiments were performed with the SmartLab at the NUCore facility with an incidence angle of 1°2θ, thickness setting of 1.5 mm, and angular range of 20 to 90°2θ for $CuBSe_2$ samples, and 20 to 60°2θ for lithiated samples, respectively under ambient conditions. The samples were prepared by drop casting the colloidal samples onto Si substrates and allowing them to dry in air. Analyses were performed using MDI-JADE, GSAS-II and Vesta.

Rietveld Refinements on XRD Data: Refinement analyses were performed with the GSAS II software[62], and crystal structures with their simulation patterns for all phase transformations were generated with Vesta.

TEM Experiments: Low-resolution TEM images were obtained using a JEOL 1400 Flash TEM at Loyola University Chicago's Biology Imaging Lab. $CuBSe_2$ samples dispersed in toluene post synthesis were initially centrifuged at 1000 rpm for 15 min and the supernatant (SPN) was isolated. Methanol was then added, and the SPN was centrifuged a second time at 10 000 rpm for 10 min. The resulting precipitate was collected and dispersed in toluene followed by sonication, and one drop of each sample was placed carefully onto fresh grids. TEM grids from Ted Pella with ultrathin-Ni 400 mesh were used. The grids were then washed in methanol and isopropyl alcohol (IPA) and left to dry under vacuum for at least 20 min. The grids were analyzed using TEM at a voltage of 80 kV and an exposure of 2000 ms × one frame with magnifications set at 8, 12, and 20 kX, respectively.

HRTEM Experiments: HRTEM experiments were performed using a JEOL ARM-200CF AC-S/TEM at the NUANCE EPIC facility. TEM grids from Ted Pella with ultrathin-Ni 400 mesh were used onto which a drop of the sample to be image was drop cast and allowed to dry under vacuum. The grids were analyzed using HRTEM at an accelerating voltage of 200 kV and an exposure of 0.5 s with magnifications set at 40, 100, 500 kX, 1 MX and 1.5 MX respectively.

DSC: DSC measurements were carried out on a PCM DSC2500 high-throughput instrument at the IMSREC facility. T-Zero aluminum pans were weighed and tared and colloidal samples of $CuBSe_2$ and lithiated versions were drop cast and allowed to dry in air. The mass of the sample was recorded each time, and the pans were added to the autosampler. DSC procedure involved an isothermal step for 1 min, ramp of 10 °C/min up to 250 °C ($CuBSe_2$), and 500 °C (lithiated versions), and a ramp of 10 °C/min down to −30 °C. This step was repeated twice consecutively.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Yunhao Xu: methodology (equal). **Niket Powar**: formal analysis (equal). **Kelly Zervos**: methodology (equal). **Isabella Campbell**: methodology (supporting). **Simran Choudhary**: methodology (supporting). **Layan Al-Khaled**: formal analysis (supporting). **Progna Banerjee**: conceptualization (lead); formal analysis (lead); investigation (lead); project administration (lead); resources (lead); supervision (lead); writing—original draft (lead); methodology (supporting); writing—review and editing (lead). **Yunhao Xu**, **Niket Powar**, and **Kelly Zervos** contributed equally to this work.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

colloidal hot injection, copper boron selenide, lithiation, nanocrystal, superionic conductor

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